

STACK OVERFLOW SOFTWARE DEVELOPMENT SURVEY

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EXECUTIVE SUMMARY

THE WEB: Still the One

Web based languages and technologies
continue to be most used and most desired

SIMPLE IS SEXY

Languages and technologies that are easy to use
(simple syntax and typing to avoid errors) are more desired by the majority

INTRODUCTION

01

STACK OVERFLOW

using every developer's Favorite website's survey to
understand trends in technology

02

PROBLEM

Software development languages and technologies
are changing quickly, in what direction are they trending?

03

QUESTIONS TO ASK

What are the technology demands of the world
and what are the demands of the developer?

04

WHAT TO LOOK FOR

Focus on what the developers work with now, what they want,
and who they are

METHODOLOGY

DATA COLLECTION & WRANGLING

Modified GITHUB Jobs API

Programming languages salaries from web scraping

Modified Stack Overflow survey results

DATA ANALYSIS & VISUALIZATION

Exploring the data through visuals

Doing analysis of the data to create meaningful insights

Creating dashboards with use of relevant data

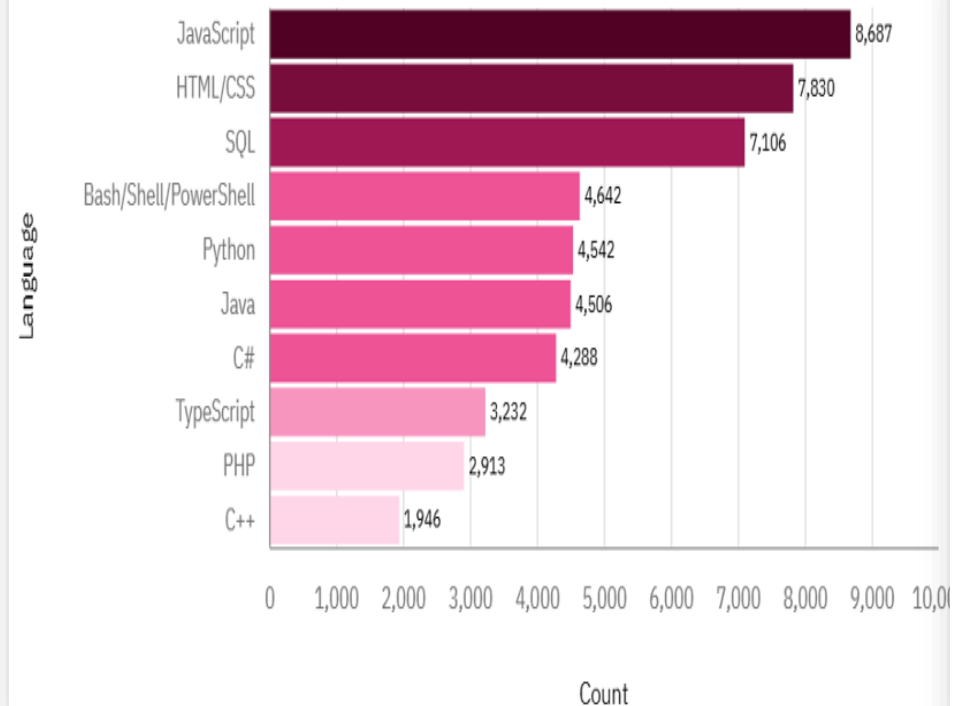
RESULTS

“The results speak for themselves.”

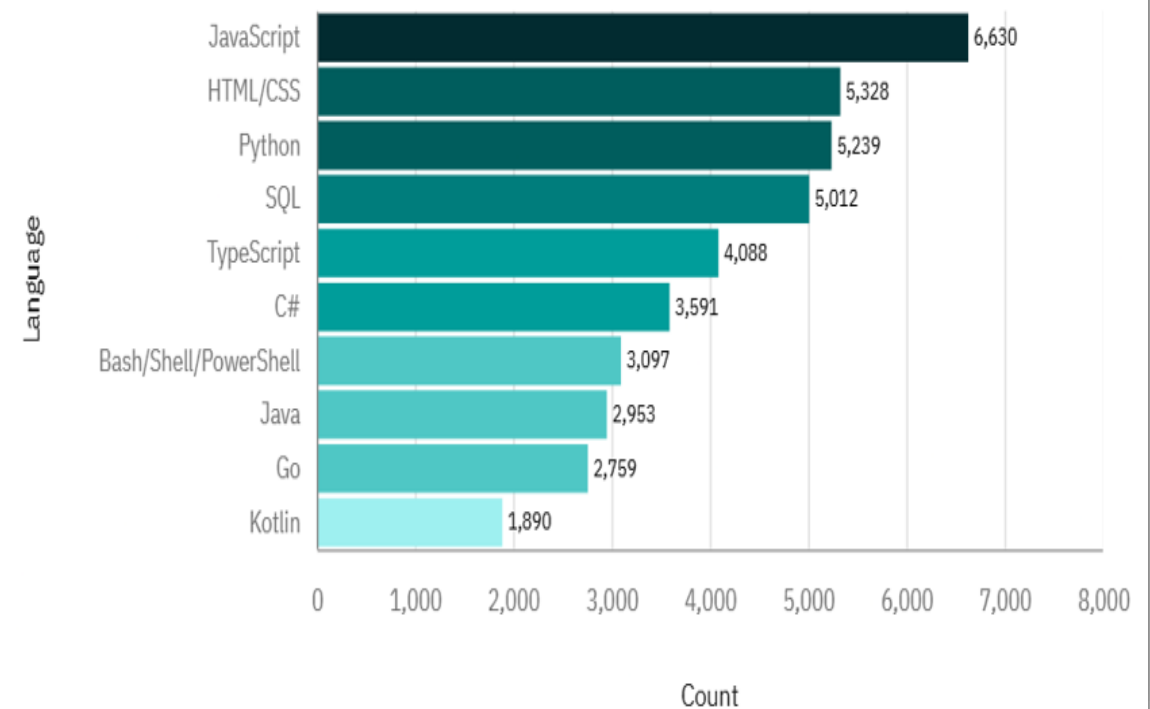
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PROGRAMMING LANGUAGE TRENDS

Top 10 Languages Worked With



Top 10 Languages Desired Next Year



PROGRAMMING LANGUAGES TRENDS- FINDINGS & IMPLICATIONS

FINDINGS

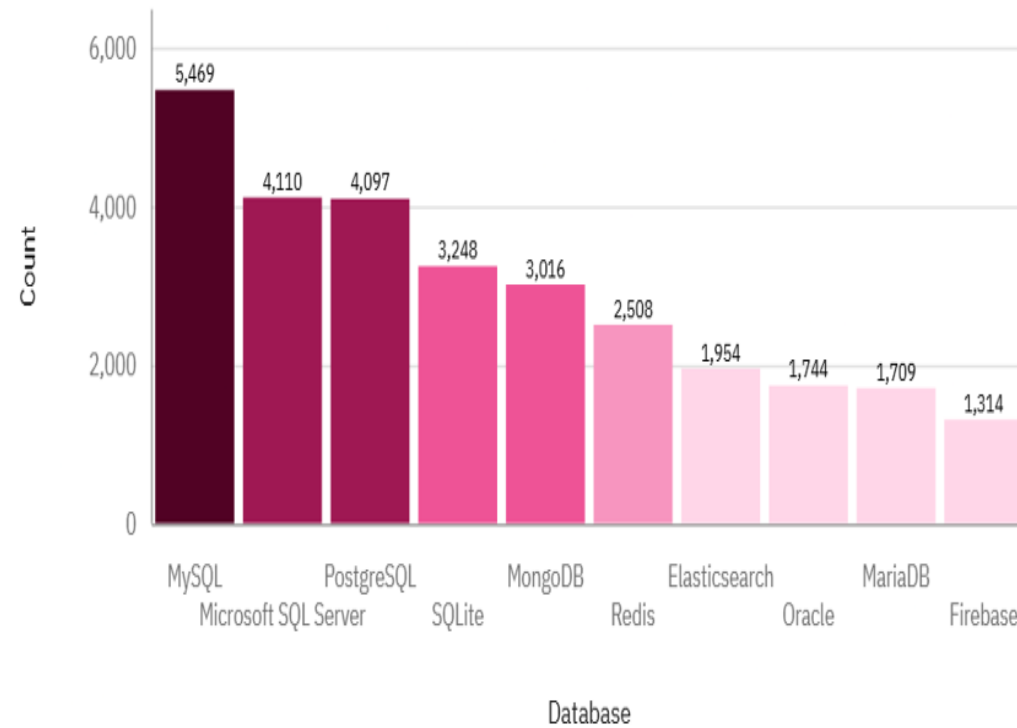
- JavaScript is the most used and most desired language
- Python demand is increasing fast
- The majority are losing interest in Bash and Shell scripting

IMPLICATIONS

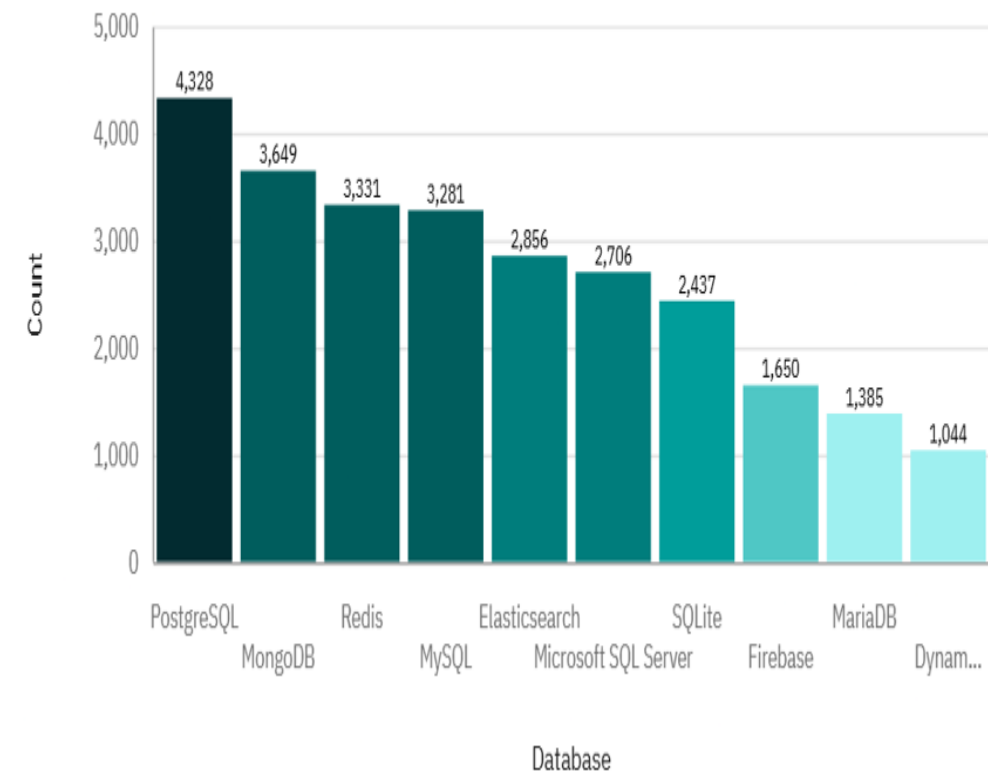
- Since the web is such a prominent aspect of the tech-world it makes sense that web languages are still at top
- JavaScript is being continually polished and made more easy to use, a large complaint is the lack of Typing.
- This could be the reason for the surge in TypeScript
- PHP, Java and C are seen as more complex and take a little more effort to work with which could be why we're seeing their decrease
- Go & Kotlin are languages that contain Typing and have simple syntax along with other features that make them easier to use than other languages mentioned earlier.

DATABASE TRENDS

Top 10 Databases Worked With



Top 10 Databases Desired Next Year



DATABASE TRENDS- FINDINDGS & IMPLICATIONS

FINDINGS

- NoSQL (Non-relational) databases are increasing in interest and demand such as MongoDB and DynamoDB
- Relational Databases are still popular
- Services are increasing in popularity Such as Elasticsearch and Firebase
- Data stores are increasing in demand such as Redis
- Oracle is not in the top 10 most desired
- Developers are losing interest in Microsoft SQL Server

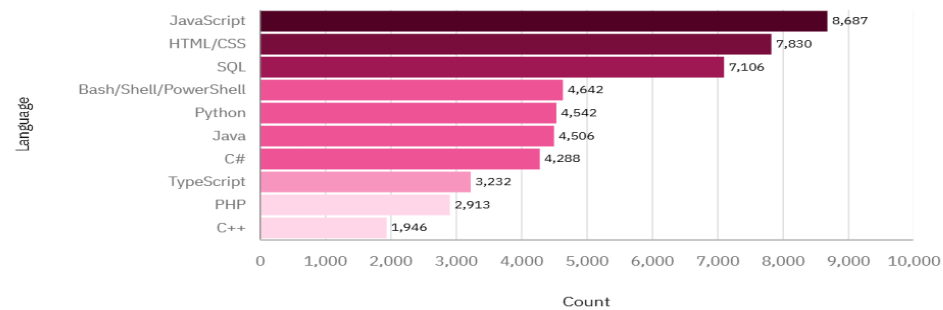
IMPLICATIONS

- PostgreSQL has more features and makes working with SQL easier, which shows a similar trend as with programming languages that simplicity and ease of use are important
- With more complex data the freedom that NoSQL provides is very alluring
- PaaS tools are gaining popularity, again the aim is to simplify the developers life
- As Data becomes more prevalent and valuable storing it becomes more important, thus explaining the increase in data stores

DASHBOARD TAB 1

Current Technology Usage

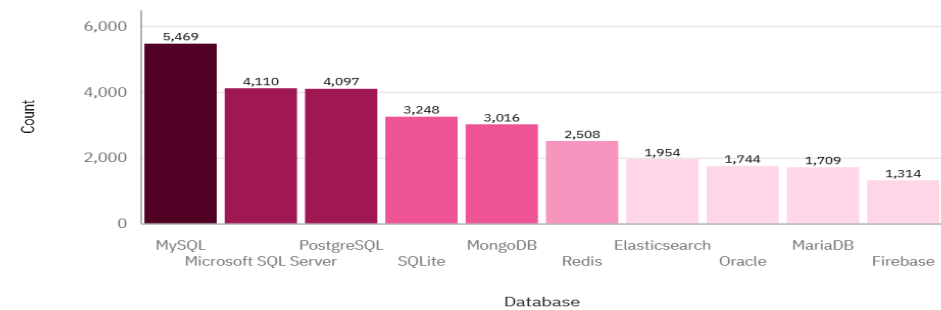
Top 10 Languages Worked With



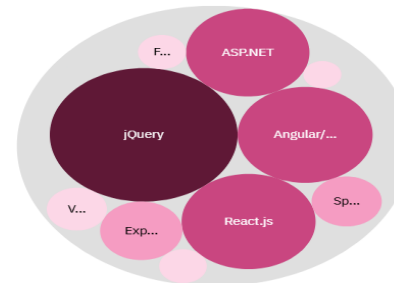
Top Platforms Worked With



Top 10 Databases Worked With



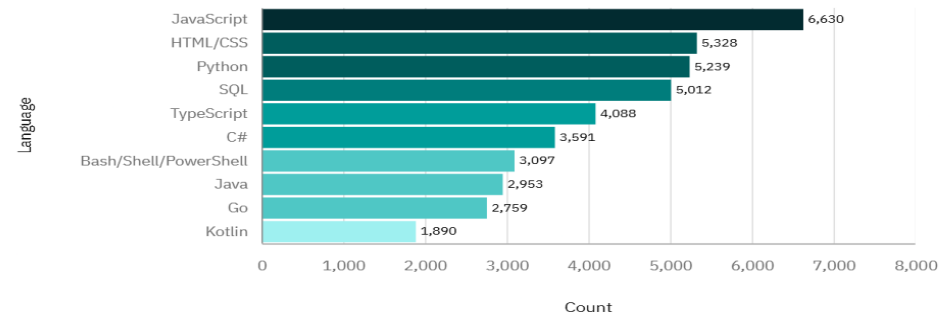
Top 10 Web Frameworks Worked With



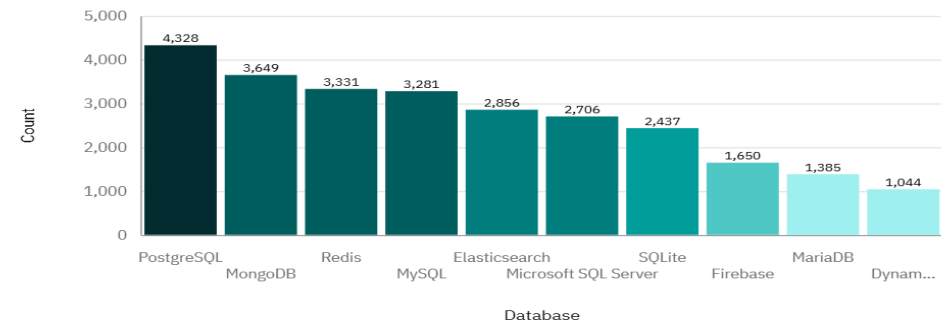
DASHBOARD TAB 2

Future Technology Trend

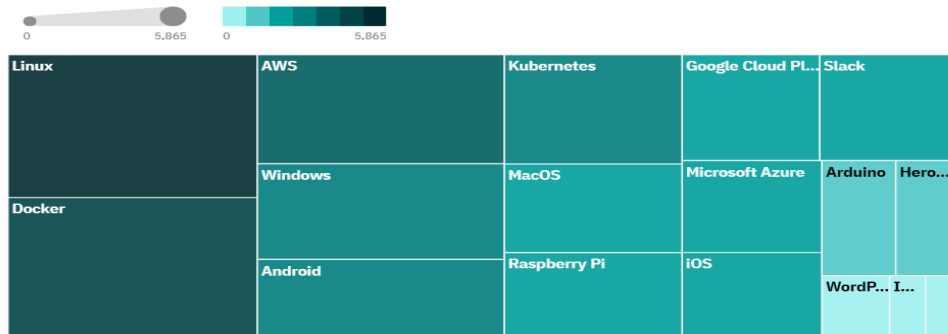
Top 10 Languages Desired Next Year



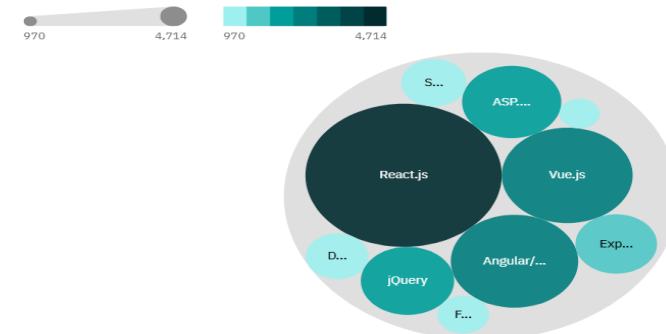
Top 10 Databases Desired Next Year



Top Platforms Desired Next Year



Top 10 Web Frameworks Desired to Work With Next Year



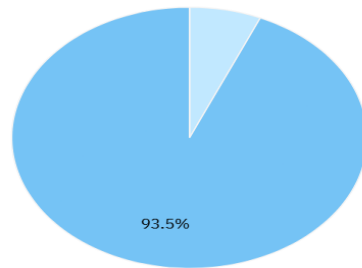
DASHBOARD TAB 3

Demographics

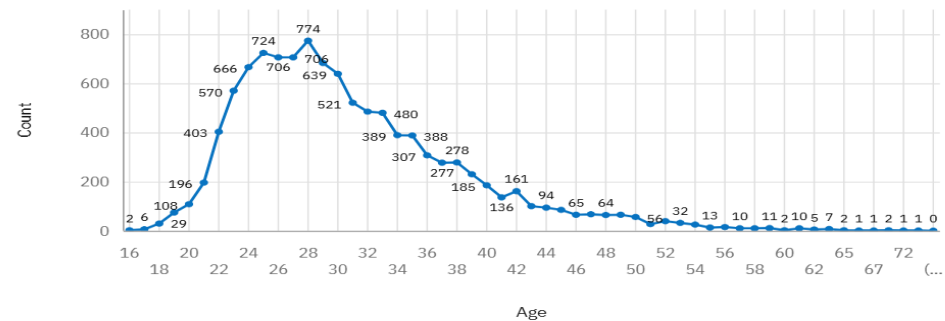
Respondent Count by Gender

Gender

Woman Man



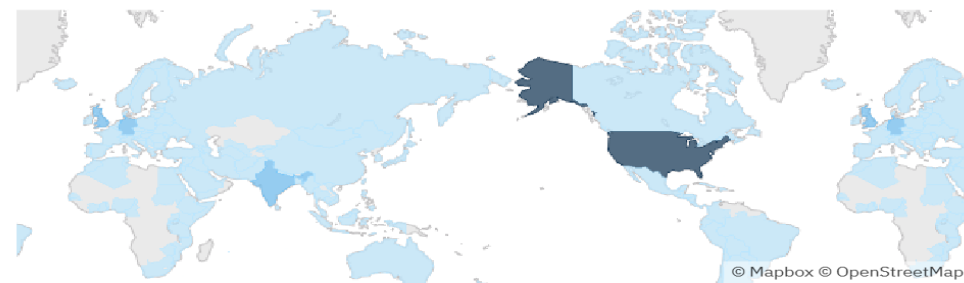
Respondent Count by Age



Respondent Count by Country

Country (Count)

1 3,058



Respondent Education Level Count by Gender

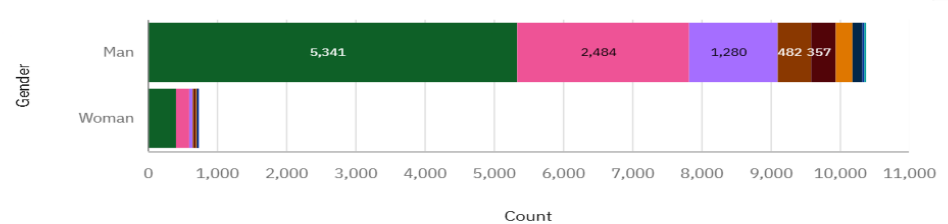
Education Level

● Bachelor's degree (BA, BS, B.Eng., etc.)

● Master's degree (MA, MS, M.Eng., MBA, etc.)

● Some college/university study without earning a degree

● Secondary school (e.g. American high school, German Realschule or Gymnasium, etc.)



DISCUSSION



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OVERALL FINDINGS & IMPLICATIONS



FINDINGS

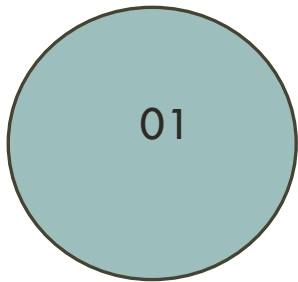
- Majority of respondents are male
- Most of the respondents were between the ages of 22-32
- Most of the respondents are from developed English speaking nations
- The most important factor for a Job is the Languages and Technologies used
- Python has one of the highest salaries and most jobs available, that could be the reason for the surge in desire

IMPLICATIONS

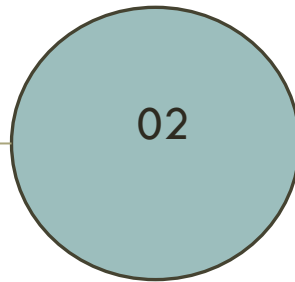
- The respondents being more on the young side could explain the tendency to want to adopt newer languages and technologies
- All the technologies in the survey are English-based, this could stem from the respondents being predominantly English
- Since developers choose jobs based on technologies used it could sway employers to embrace those technologies in use to draw in talent

CONCLUSION

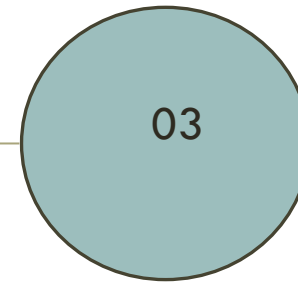
Developers are looking for 3 main factors when choosing languages and technologies



Languages that are
Syntactically simple



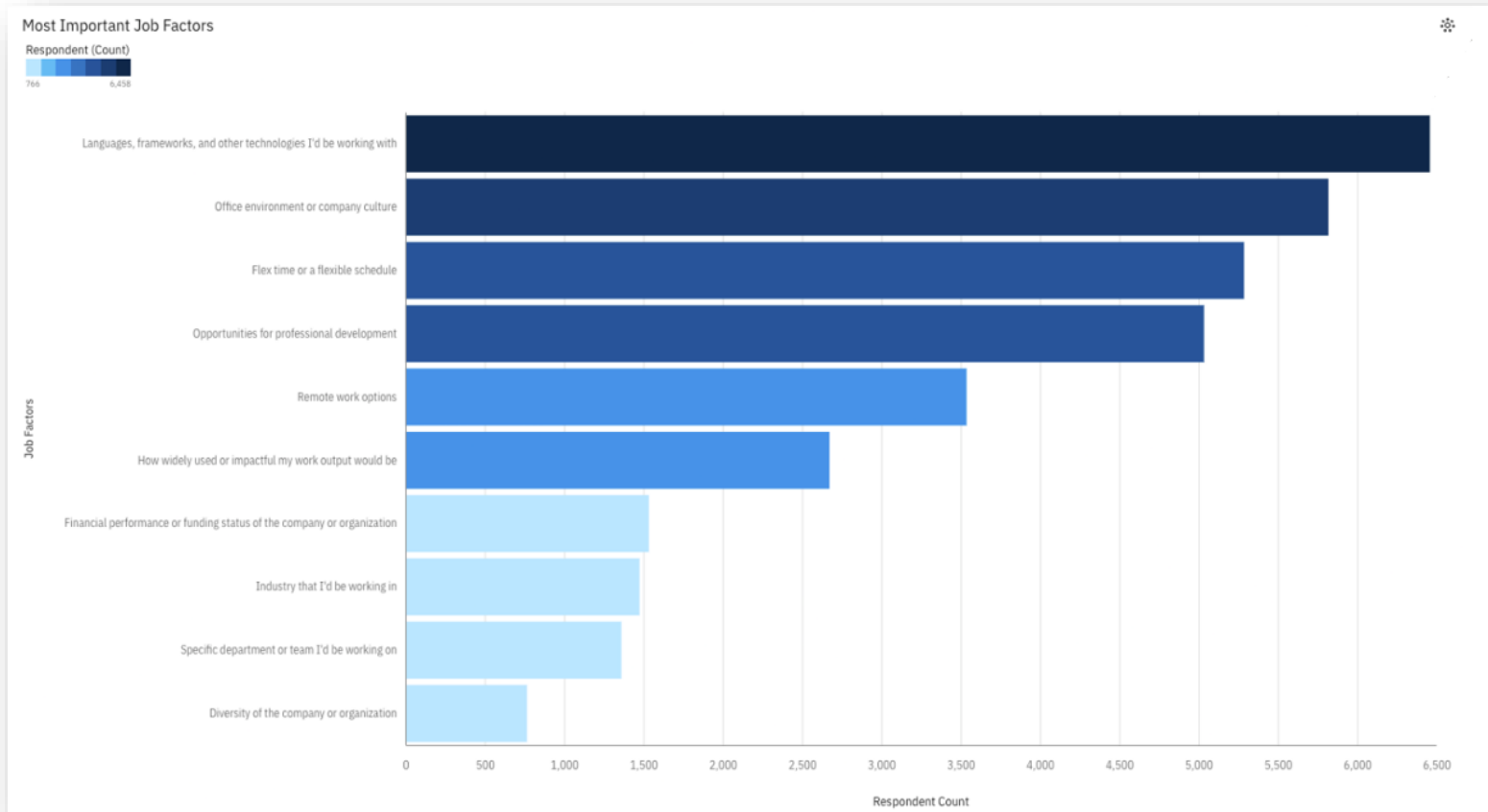
Tools that require less
overhead and are
more declarative
rather than imperative



Tools that are less
error-prone which
can be
accomplished by
Typing

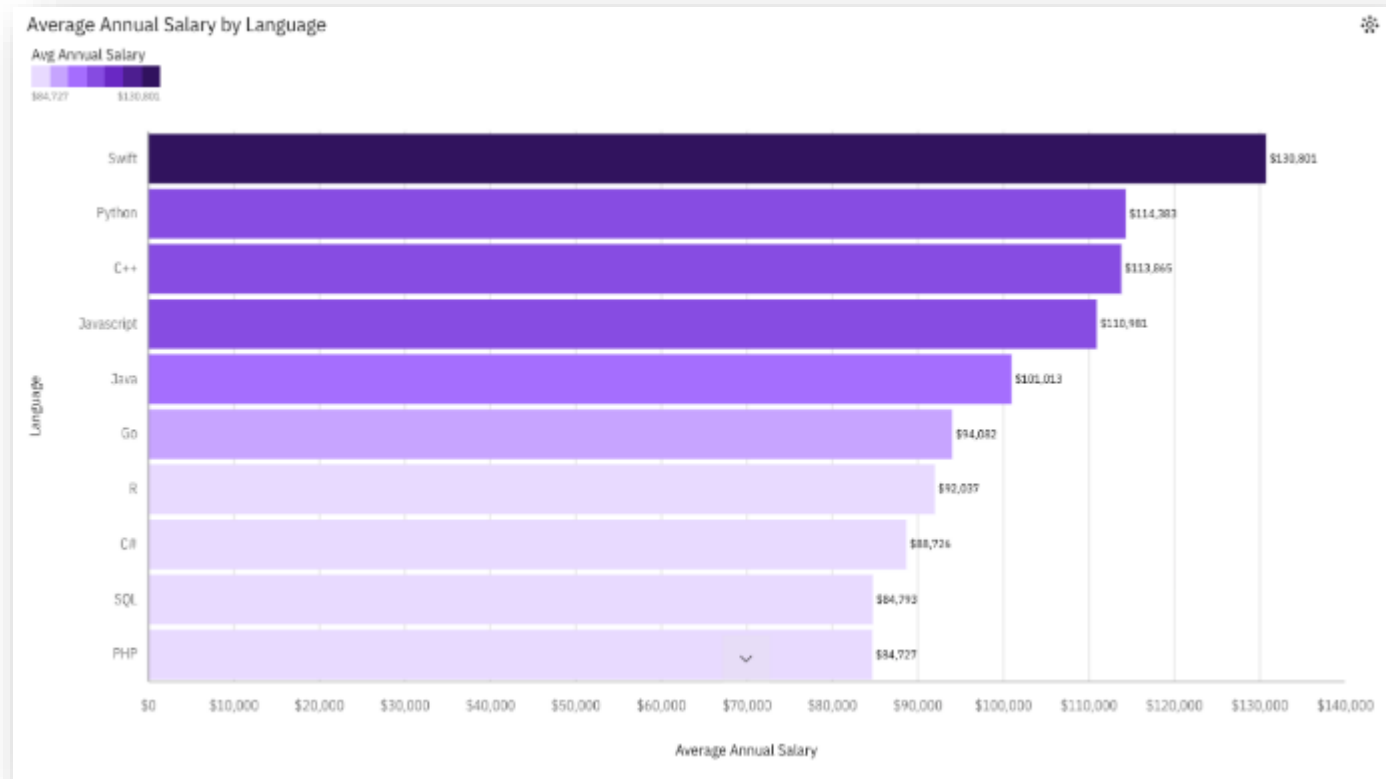
APPENDIX

TOP FACTOR

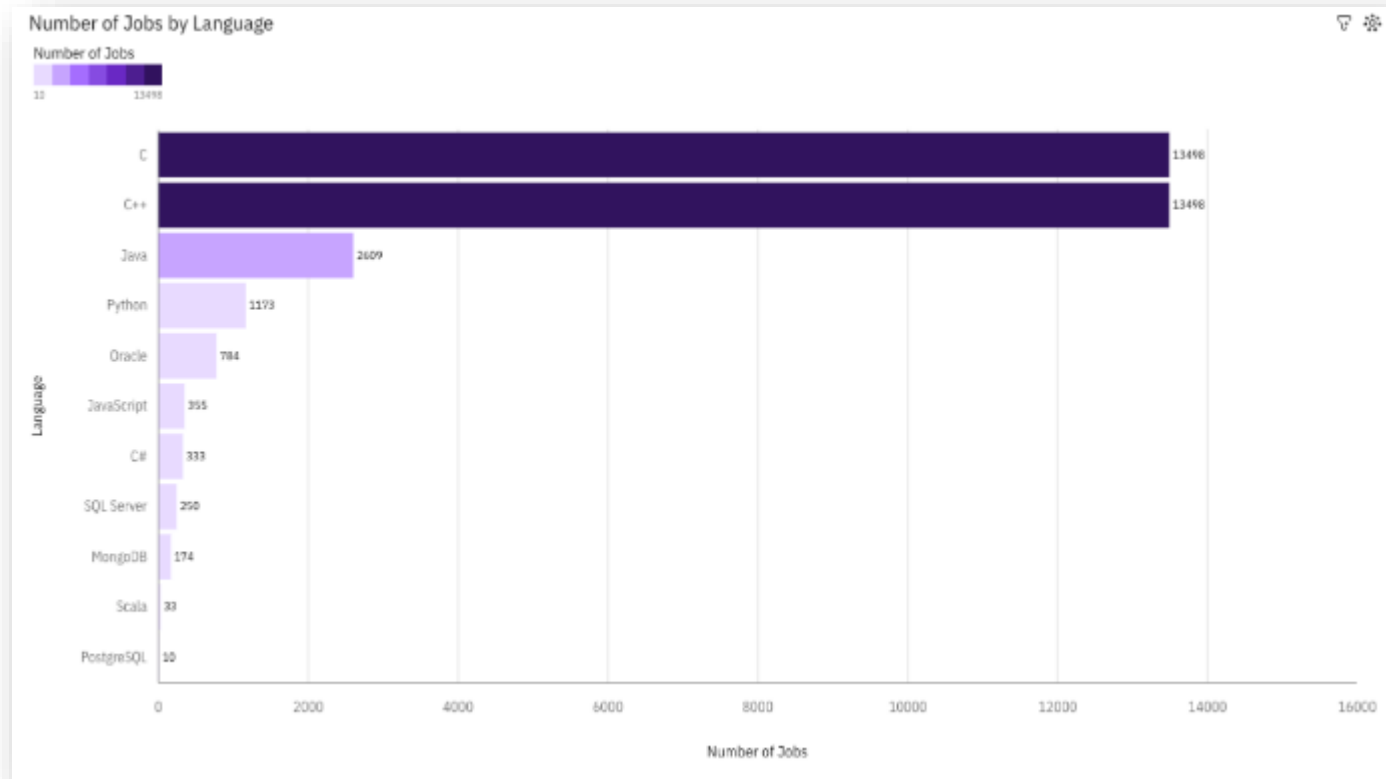


Most important Job factor is the coding technologies Respondents work with

POPULAR LANGUAGES



JOB POSTINGS



SOURCES

stack overflow annual developer survey 2019 from

<https://survey.stackoverflow.co/2019>